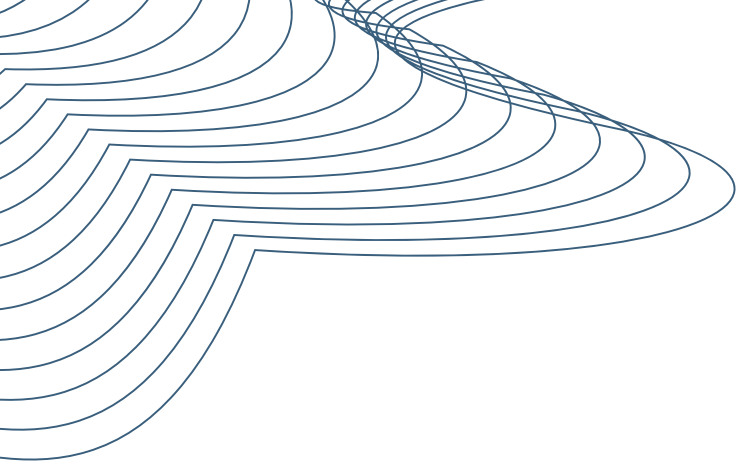
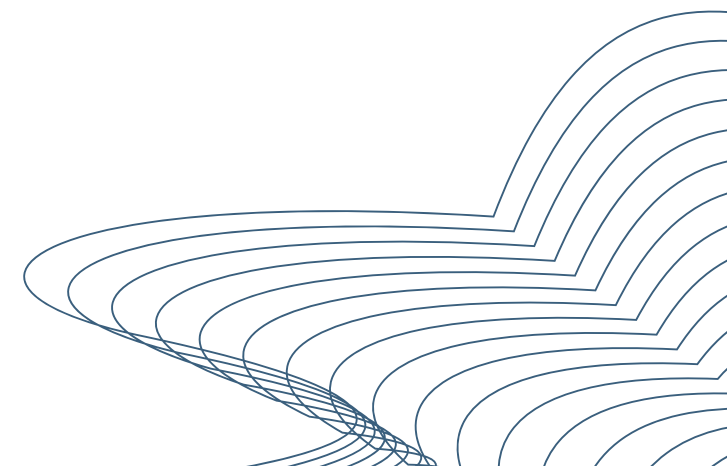




Automate Terraform with CI/CD to automate infrastructure provisioning

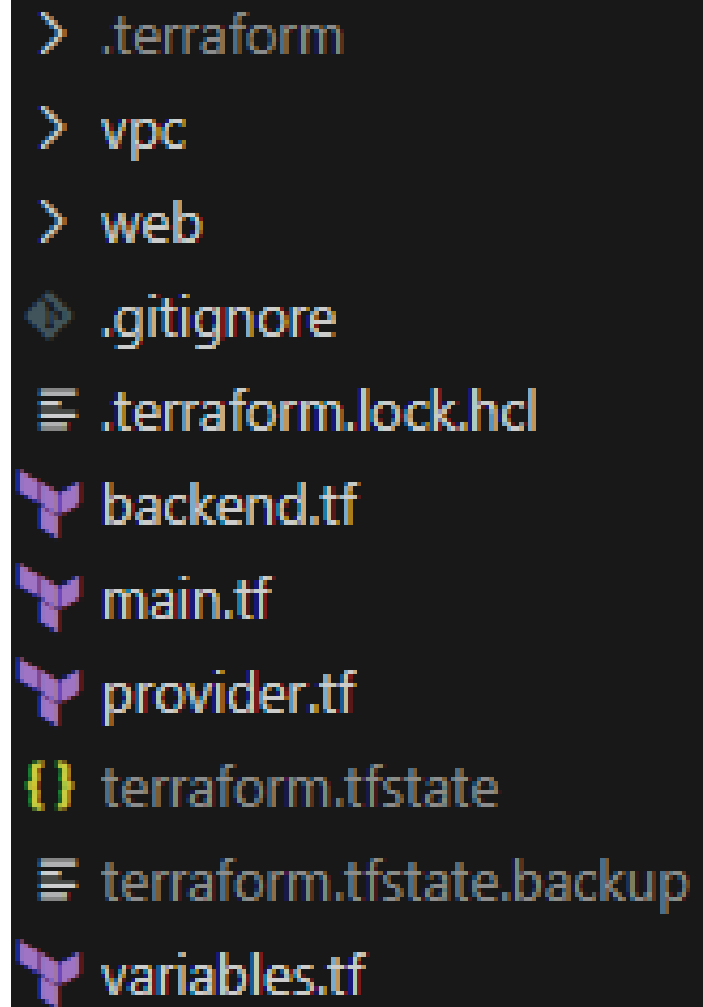


Provider

aws

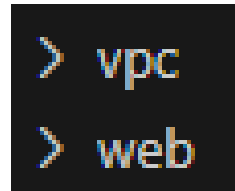


Files Used



- > .terraform
- > vpc
- > web
- ◆ .gitignore
- ≡ .terraform.lock.hcl
- 📄 backend.tf
- 📄 main.tf
- 📄 provider.tf
- { } terraform.tfstate
- ≡ terraform.tfstate.backup
- 📄 variables.tf

Module Block

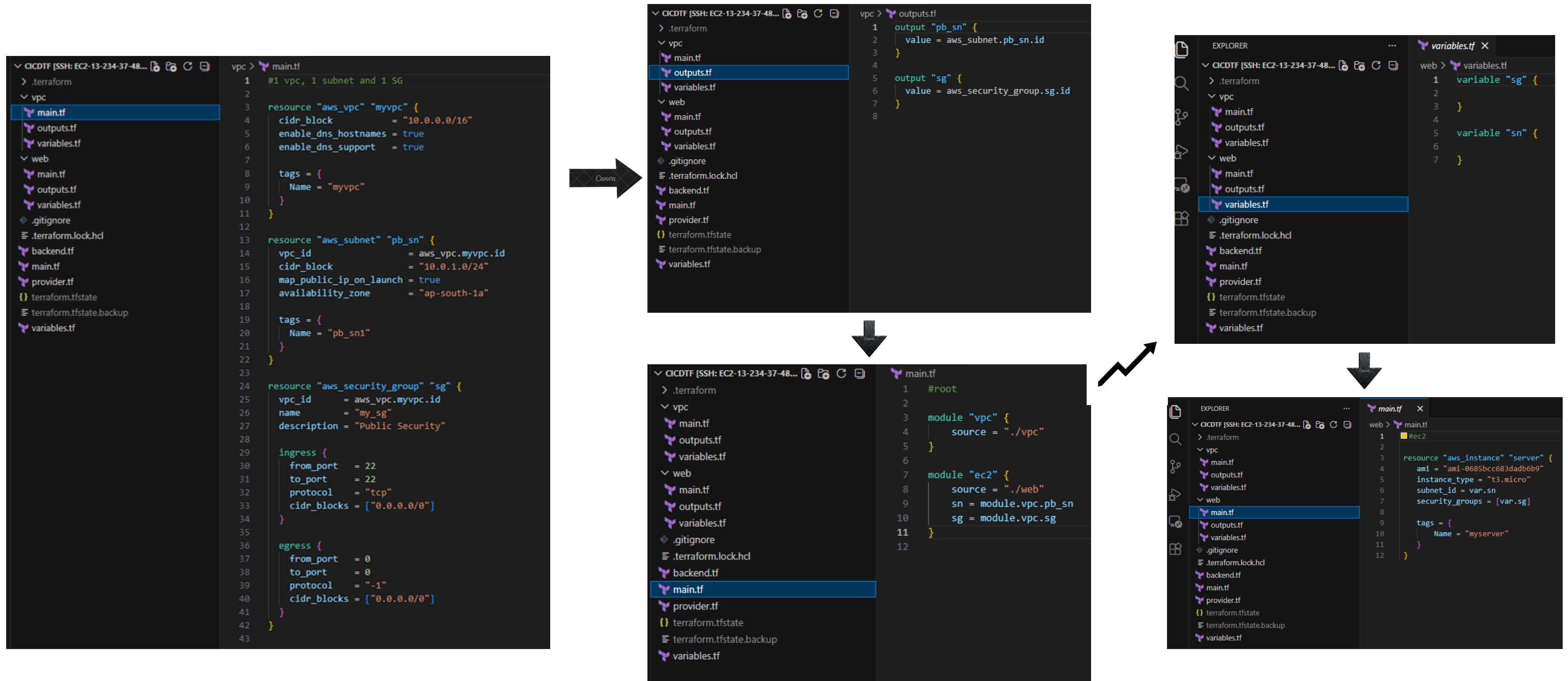


Module

1. module is a reusable unit which distributes our infracture so that we can use it unless writing it.
2. module takes parameters/inputs so that modules works according to need

Flow

VPC Module → main.tf (vpc module) → Generates id of pb_sn & sg resource → Values comes in output.tf (vpc module of pb_sn & sg) → Same values goes to main.tf (root) → now module "ec2" have same value in sn & sg (root main.tf) → Same values goes to sg & sn of variables.tf (web module) → that sn & sg value we are using in web (main.tf)



Objects (1)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☐	Name	Type	Last modified	Size	Storage class
☐	state	-	June 5, 2026, 10:53:37 (UTC+05:30)	10.7 KB	Standard

S3 Bucket

DynamoDB

☐	mystatebucketterraform2332	Asia Pacific (Mumbai) ap-south-1	June 1, 2026, 10:48:30 (UTC+05:30)
-----------------------	--	----------------------------------	------------------------------------

☐	backend	Active	LockID (S)	-	0	0	Off	☆	On-demand	On-demand	79 bytes
--------------------------	-------------------------	--------	------------	---	---	---	-----	---	-----------	-----------	----------

Use: use for state locking in dynamoDB is when someone change terraform state file so at that time other can't update the file

CI/CD Configuration Script

 .gitlab-ci.yml

Find file

Edit



Update .gitlab-ci.yml file
Harshal Jain authored 2 days ago



e3e347e4



History

 .gitlab-ci.yml 978 B

Blame



```
1 image:
2   name: registry.gitlab.com/gitlab-org/gitlab-build-images:terraform
3   entrypoint:
4     - '/usr/bin/env'
5     - 'PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin'
6
7 variables:
8   AWS_ACCESS_KEY_ID: ${MY_AWS_ACCESS_KEY}
9   AWS_SECRET_ACCESS_KEY: ${MY_SECRET_KEY}
10  AWS_DEFAULT_REGION: "ap-south-1"
11
12 before_script:
13   - terraform --version
14   - terraform init
15
16 stages:
17   - validate
18   - plan
19   - apply
20   - destroy
21
22
23 validate:
24   stage: validate
25   script:
26     - terraform validate
27
```

27

28

plan:

29

stage: plan

30

dependencies:

31

- validate

32

script:

33

- terraform plan -out="planfile"

34

artifacts:

35

paths:

36

- planfile

37

38

apply:

39

stage: apply

40

dependencies:

41

- plan

42

script:

43

- terraform apply -input=false "planfile"

44

when: manual

45

46

destroy:

47

stage: destroy

48

script:

49

- terraform destroy -auto-approve

50

when: manual

Artifacts

We store the result in one file, We use the same file in otherjob

CI/CD Variables </> 2				Reveal values	Add variable
Key ↑	Value	Environments	Actions		
MY_AWS_ACCESS_KEY  Protected Masked	 	All (default) 			
MY_SECRET_KEY  Protected Masked	 	All (default) 			

```
-  
7  variables:  
8      AWS_ACCESS_KEY_ID: ${MY_AWS_ACCESS_KEY}  
9      AWS_SECRET_ACCESS_KEY: ${MY_SECRET_KEY}  
10     AWS_DEFAULT_REGION: "ap-south-1"
```

Pipelines

Status	Pipeline	Created by	Stages	Actions
✓ Passed 🕒 00:02:36 📅 2 days ago	#2578352204 Update .gitlab-ci.yml file 📁 main ➡ e3e347e4 📁 📁 latest 📁 branch		✓ ✓ ✓ ⌵ Stage: apply ✓ apply ▶	▶ ⌵ 📄 ⌵
✓ Passed 🕒 00:02:36 📅 2 days ago	#2578352204 Update .gitlab-ci.yml file 📁 main ➡ e3e347e4 📁 📁 latest 📁 branch		✓ ✓ ✓ ⌵ destroy: skipped Stage: destroy ⊖ destroy ▶	▶ ⌵ 📄 ⌵

EC2 Server

myserver	i-Oe9bbe510f11cxb6/9	⏸ Stopped 🔍 🔍	t3.micro	-	ap-south-1a	-	-	-	-
----------	----------------------	---------------	----------	---	-------------	---	---	---	---

THANK YOU